

Ictal Patterns (Electrographic Seizures)

Ictal patterns are repetitive abnormalities that electrographically represent ongoing seizure activity.

The identification of ictal patterns is largely based on their clinical accompaniments and electrographic features. Evidence of regional hypermetabolism (PET) or increased blood flow (SPECT) on nuclear imaging techniques may also suggest an ictal focus. The rationale for aggressive treatment of these patterns should be based on the possibility of clinical improvement with the resolution of the abnormality [1,2].

Clinically, ictal patterns may be accompanied by subtle focal manifestations such as clonic activity or altered mentation (nonconvulsive seizures).

Electrographically, the hallmark of an ictal pattern is “evolution.” The presence of “plus features” also renders the pattern more ictal appearing than the same pattern without the plus feature (also known as a plus term) [3].

Evolution

An evolving pattern shows at least two sequential electrographic changes in frequency, morphology, or location. Changes in amplitude or sharpness alone do not qualify as evolution.

The individual change should persist for at least three cycles (three times) and not exceed 5 min in duration to qualify (e.g., a change in frequency as 3 Hz for 1 min to 2 Hz for 8 min to 1 Hz for 2 min doesn't qualify):

- At least two sequential frequency changes, which may be either increases (crescendo) or decreases (decrescendo) of more than 0.5 Hz each, will qualify.
- At least two sequential morphological changes to a new morphology will also qualify.
- Spreading in or out of at least two different electrode locations will also qualify.

If the pattern changes but doesn't meet the criteria for evolution, it is called a “fluctuating pattern.” Static patterns refer to those that do not evolve or fluctuate [3]. Figure 16.1 shows an evolution in frequency, morphology, and spread.

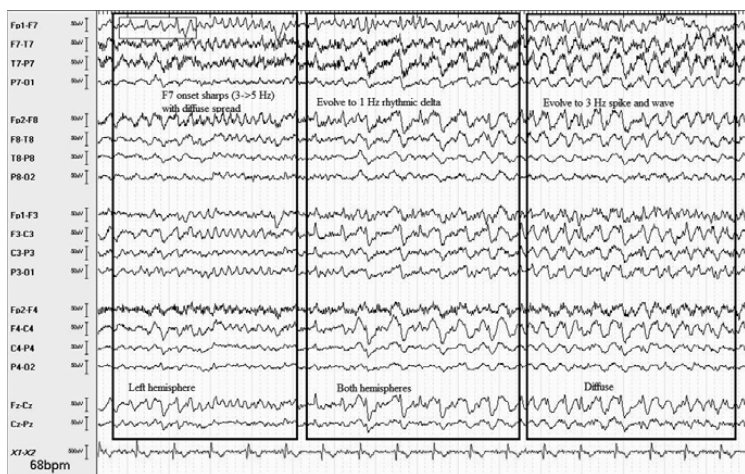


Figure 16.1 Seventy-seven-year-old with altered mentation, onset is characterized by low amplitude LPDs (3 Hz) in the Fp1 channel that initially evolves to diffuse low amplitude sharply contoured theta (5–6 Hz) [frequency] followed by frontal-predominant higher amplitude rhythmic delta (1 Hz) with embedded spikes [morphology] that spread to both hemispheres [location] and then increase in frequency to frontal-predominant generalized spike-wave discharges (2–3 Hz) before abruptly terminating (time base 15 s). Therefore, in this example, the original waveform evolves in frequency, morphology, and location (spread) constituting an electrographic seizure.

“Plus” Term

As discussed before, the presence of a “plus” feature also renders the repetitive pattern more ictal in appearance. LPD with plus features (usually superimposed low-amplitude fast activity or “F”) are much more likely to be associated with epileptic seizures compared to those without plus features (as high as 74% based on a single study) [3,4].

Use the following “plus terms” with the combination of main terms as follows:

- + F: for superimposed low amplitude fast activity (e.g., GPD+F). See Figure 16.2.
- + R: for superimposed rhythmic delta activity (e.g., GPD+R). See Figure 16.3.
- + S: for superimposed sharp waves or spikes (e.g., Left LRDA+S). See Figure 16.4.

The term plus fast (+F) may be applied to both periodic discharges and rhythmic delta activity. Plus sharp (+S) is not applicable to periodic discharges

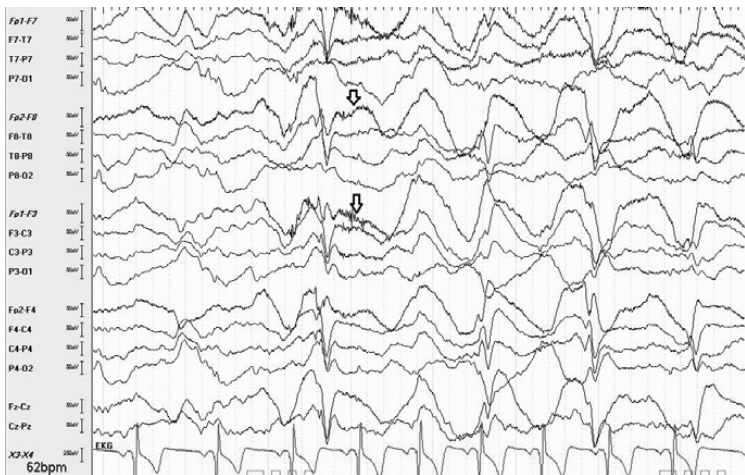


Figure 16.2 Thirty-three-year-old man with generalized epilepsy, GPDs with plus F features (fast). Brush-like low-amplitude rhythmic fast activity with a frontal predominance overrides the periodic discharge.

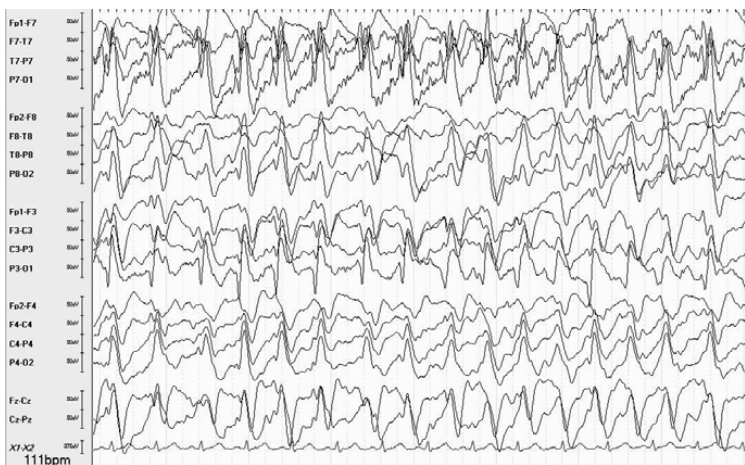


Figure 16.3 Same patient as in Figure 16.2, GPDs with plus R features (rhythmic) seen during a seizure. Spikes should not be time locked to slow waves, else they qualify as spike and wave (SW) instead of PD plus S.

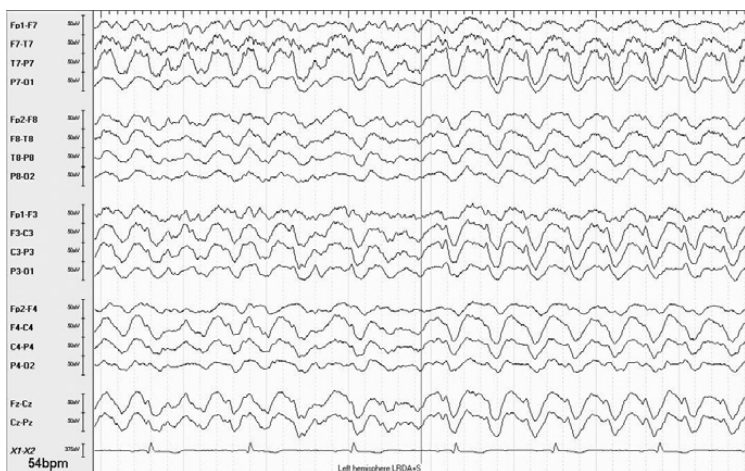


Figure 16.4 Same patient as in Figure 16.1, left hemispheric LRDA with plus S features (sharps). Low-amplitude sharps override the rhythmic waveforms.

as they are inherently sharp; conversely, plus rhythmic (+R) is not applicable to rhythmic delta activity as this is inherently rhythmic. Generally, plus features do not apply to spike and wave complexes.

Electrographic Seizures

Electrographic seizures (nonconvulsive or subclinical seizures) are epileptic seizures diagnosed on EEG alone. They occur without obvious clinical accompaniments.

Conventionally, an electrographic seizure has a clear start (onset) followed by evolution and termination (offset) with a duration that by convention is 10 s or more. Voltage attenuation (suppression) or slowing may be seen immediately postseizure. The seizure pattern may consist of evolving rhythmic activity (alpha, theta, or delta range), sharp/spike waves, fast activity (beta range), periods of attenuation, or combinations of these.

Heralding discharges (such as spikes) may occur immediately prior to onset and periodic discharges are common post offset (e.g., post ictal LPDs).

Electrocardiographic abnormalities such as tachycardia and less commonly bradycardia or asystole may be seen on the EKG channel during the seizure. In neonates and young children, ictal apneas may also occur.

Electrographic seizures are defined as patterns of 10 s or more in duration with on any one of three criteria (Salzburg criteria) [3,5,6]:

1. repetitive epileptic discharges of frequency greater than 2.5 Hz; OR,
2. repetitive epileptic discharges or rhythmic slowing of frequency less than 2.5 Hz AND shows at least one of the following features:
 - a. clinical accompaniments (may be subtle); these are referred to as “electroclinical” seizures.
 - b. spatiotemporal evolution as described above.
 - c. significant clinical and EEG improvement after a trial of antiepileptic medication (EEG improvement as described before must not be limited to the suppression of discharges, but emergence of previously absent normal features such as a posterior dominant rhythm).

Figures 16.5 to 16.11 show the sequential progression of atypical electrographic seizure from onset to termination.

The reader should note that the “10 s” rule is an adopted convention. Electroclinical seizures (i.e., those with a definite clinical correlate time locked to an EEG pattern, such as face or limb twitching) are classified seizures even if the accompanying electrographic pattern may be less than 10 s.

Clinical Implications of Nonconvulsive Seizures

1. Seizures occur in 8%–20% of critically ill patients with altered mentation; they are almost always nonconvulsive. These patients may not have acute brain injury or prior epilepsy [7].
2. Nonconvulsive seizures (NCS) or nonconvulsive status epilepticus (NCSE) may lead to persistently abnormal mentation despite apparent successful treatment of convulsive status epilepticus (CSE). During 24 hours of continuous EEG monitoring after CSE, NCS was recorded in 48% and NCSE in 14% in one large study [8].
3. Continuous EEG monitoring is the preferred method of diagnosis for nonconvulsive seizures as there can be several hours of delay before the first seizure is recorded. Typical 30- to 60-min recordings identify NCS in only 45%–58% of patients in whom seizures are eventually recorded [9].
4. The duration of continuous EEG monitoring should be tailored to the individual patient. Recording for at least 24 hours is recommended, and this should be extended to at least 48–72 hours in high-risk patients (those who are comatose, have prior seizures, are pharmacologically sedated, are undergoing antiepileptic drug withdrawal, or have epileptogenic abnormalities such as periodic discharges on EEG). Noncomatose

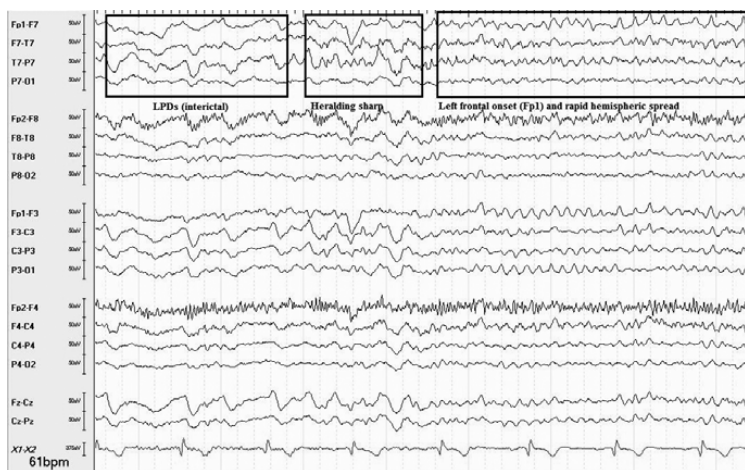


Figure 16.5 Same patient as in Figure 16.1; this electrographic seizure onset is characterized by a sharp theta activity (5–6 Hz) in the Fp1 channel followed by rapid left hemispheric spread. A heralding sharp wave at Fp1 precedes the onset.

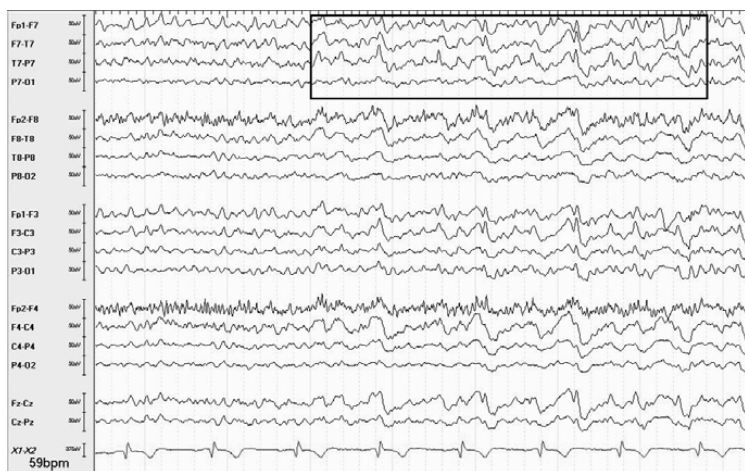


Figure 16.6 Sharp theta seen in Figure 16.5 slows to delta frequency (1–3 Hz) with emergence of embedded left hemispheric LPDs (1 Hz) with wide fields as the seizure continues.

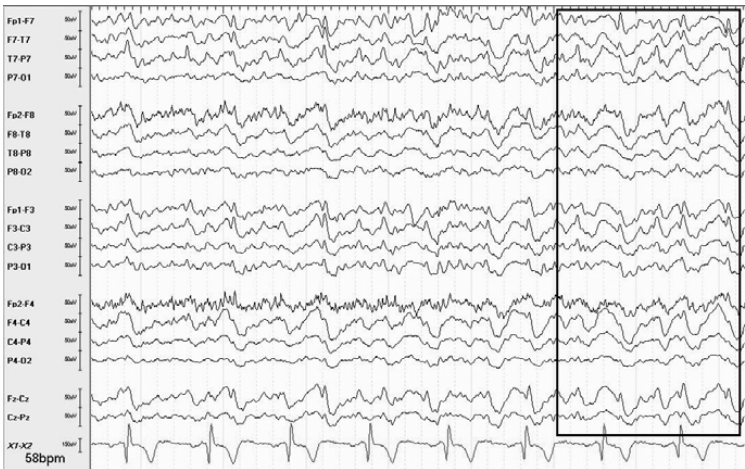


Figure 16.7 LPDs seen in Figure 16.6 become rhythmic and increase in frequency, transforming into 3 Hz LRDA (left hemisphere) with embedded sharps (LRDA plus S) as the seizure continues. This pattern has a wide bilateral field (secondary generalization).

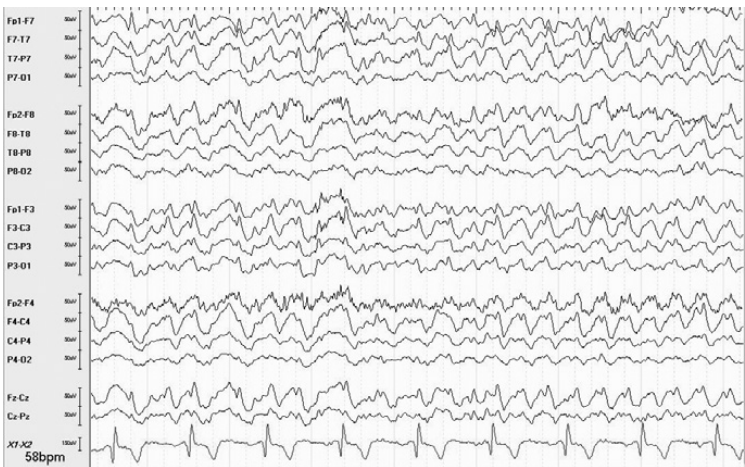


Figure 16.8 Approximately 3 Hz LRDA plus S seen in Figure 16.7 evolves in morphology to a lower amplitude and sharps become less prominent.

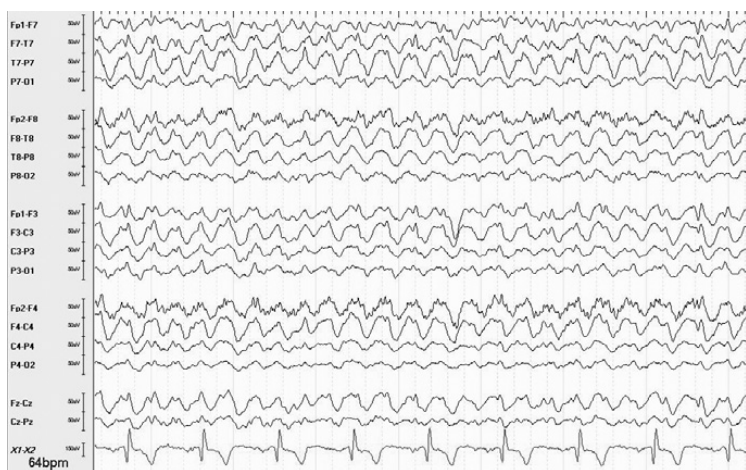


Figure 16.9 Lateralized rhythmic delta plus S seen in Figures 16.7 and 16.8 continues for about 7–8 s during the seizure.

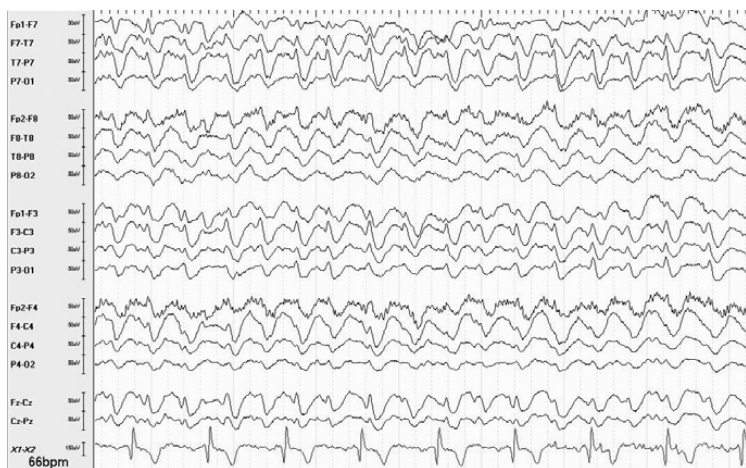


Figure 16.10 LRDA plus S seen in Figure 16.9 slightly decreases in frequency during this part of the seizure.

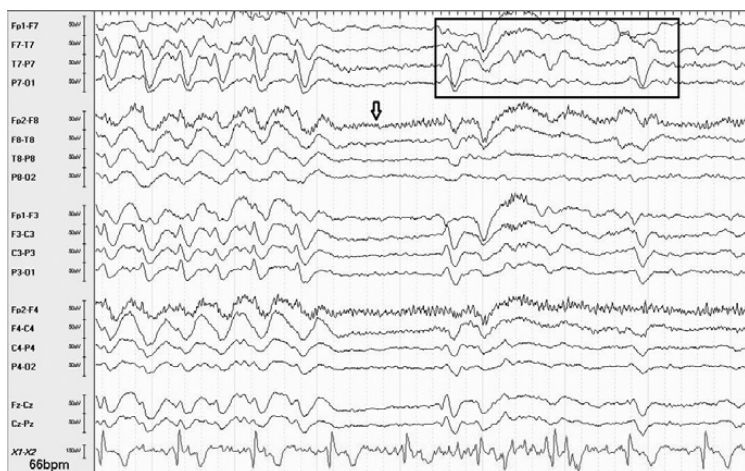


Figure 16.11 The seizure abruptly terminates, followed by a brief period of generalized suppression (postictal generalized suppression) and reemergence of left hemispheric LPDs (post ictal LPDs).

patients without prior seizures or early epileptogenic abnormalities (within 30–120 mins on EEG) have a very low risk of seizures (<5%) based on multiple studies [10,11].

5. Both NCS and NCSE are associated with increased mortality based on multiple studies, with the underlying etiology being the key prognostic factor [12,13].
6. Longer duration and greater delay in the diagnosis of NCS may contribute to worse outcomes. There is growing evidence to suggest that NCS may be associated with neuronal injury [14].

Brief Potentially Ictal Rhythmic Discharges (BIRDs)

These are typically focal sharply contoured bursts of less than 10 s duration that may show evolution or epileptic discharges. These are occasionally seen in critically ill adults with altered mentation and underlying structural injuries (e.g., strokes or tumors). Those with BIRDs are more likely to have seizures, and some may have only nonconvulsive seizures (epileptogenic). Their prognostic significance is not known [15]. Figure 16.12 shows an example of a BIRD.

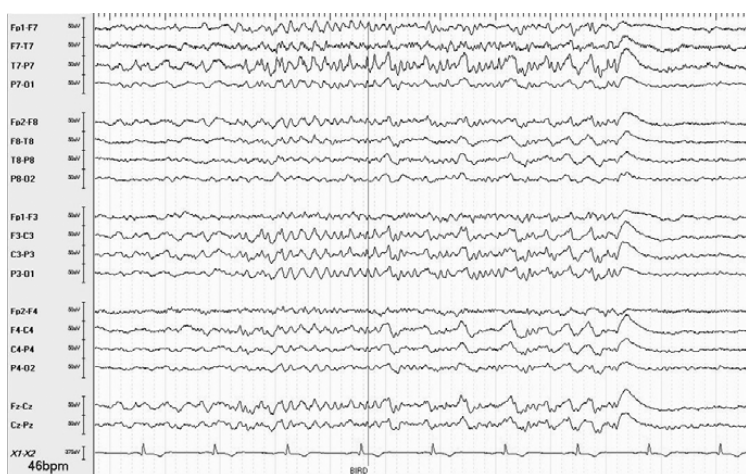


Figure 16.12 Same patient as in Figure 16.1, a brief potentially ictal rhythmic discharge (BIRD) seen during treatment of nonconvulsive status epilepticus (time base 10 s).

Chapter Summary

1. Ictal patterns represent ongoing electrographic seizures. They are recognized by their clinical accompaniments (may be subtle) and electrographic features.
2. Evolution is the electrographic hallmark of an ictal pattern, plus features (such as overriding fast activity) also render a pattern more ictal in appearance.
3. Electrographic seizures will typically have a clear onset, evolution, and offset. Conventionally, they should last for 10 s or more.
4. Electrographic seizures are common after convulsive seizures, in acute brain injury, and in critically ill patients with altered mentation.
5. Continuous EEG monitoring is the preferred method of diagnosing electrographic seizures. Typically, a duration of 24 hours is sufficient, but this should be extended in certain high-risk populations.

References

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